

Estimating the Health Impact of Achieving SEAHEARTS 2030 Milestones

University of Washington · WHO South-East Asia Regional Office

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Estimating Health Gains: Country baseline-to-target

Quantifying the health payoff of the SEAHEARTS 2030 milestones

Project aim

Quantify the potential health gains from **progress toward** and **full achievement** of the SEAHEARTS 2030 milestones — to support **country advocacy** and **regional decision-making**.

Three headline outcomes

- **Deaths averted**
- **DALYs averted** (disability-adjusted life-years)
- **% reduction in premature NCD mortality**

Coverage

Estimated for each of the **10 SEAR Member States** in the milestone tables, then aggregated to the **WHO South-East Asia Region**.

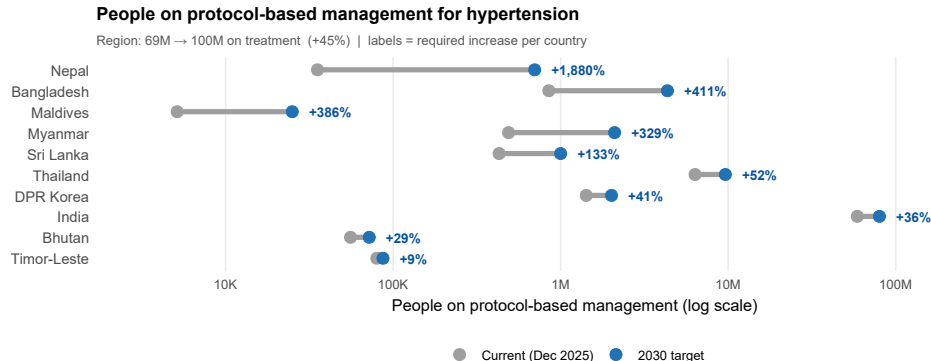
Bangladesh · Bhutan · DPR Korea · India · Maldives · Myanmar · Nepal · Sri Lanka · Thailand · Timor-Leste

SEAHEARTS 2030: five milestones across treatment and policy

Milestone area	Regional target by 2030
Hypertension	100M people on protocol-based management, incl. 60M with controlled blood pressure
Diabetes	70M people on protocol-based management, incl. 35M with good glycaemic control
Tobacco (MPOWER)	1 billion people covered by ≥ 3 MPOWER measures at the highest level of achievement
Sodium (SHAKE)	1 billion people covered by ≥ 2 additional interventions to cut mean population salt/sodium intake
Trans-fat (REPLACE)	1 billion people protected via best-practice, less-restrictive or complementary trans-fat policies

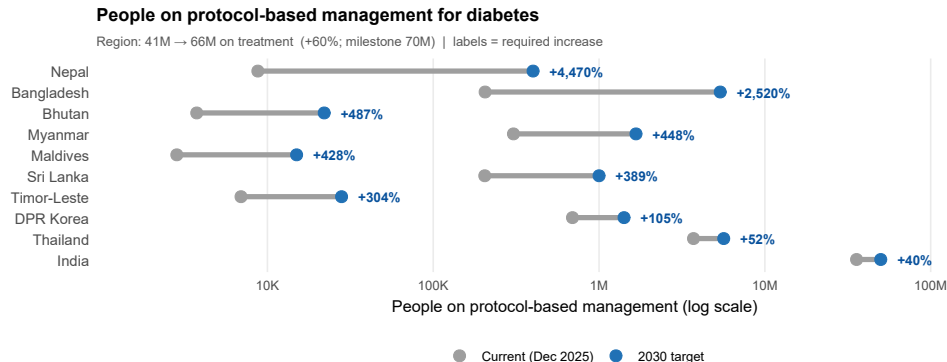
Extended and expanded regional milestones, WHO SEA Regional Committee decision SEA/RC78(7) (World Health Organization Regional Office for South-East Asia 2025); SEAHEARTS resolution SEA/RC76/R5 (World Health Organization Regional Office for South-East Asia 2023).

Hypertension: reaching the target means a step-change in treatment



Largest relative scale-up: Nepal, Bangladesh, Maldives, Myanmar ($\$ > 3\times$). **Largest absolute:** *India* ($\approx \$59\text{M} \rightarrow 80\text{M}$). Required increase = $100 \times (\text{target} - \text{baseline}) / \text{baseline}$; values are people (counts), so this is a *relative* increase, not a percentage-point change. Source: SEAHEARTS milestone tables (project document).

Diabetes: the steepest scale-up of all



Treatment coverage starts from a lower base than hypertension, so the required increases are even steeper: **Nepal** ($\approx 46\times$) and **Bangladesh** ($\approx 26\times$) stand out. Country targets sum to $\approx \$66\text{M}$; the regional milestone is 70M on management. Required increase = $100 \times (\text{target} - \text{baseline}) / \text{baseline} (\text{counts})$. Source: SEAHEARTS milestone tables (project document).

Analysis Plan

Two distinct questions — not just two time horizons

	Estimation 1 — Progress (2025→2030)	Estimation 2 — Milestones (→2040)
Policy question	What has <i>progress so far</i> already delivered?	What would <i>full achievement</i> deliver?
Scale-up modelled	Coverage <i>actually reached</i> by countries in 2025	<i>Full</i> national & regional 2030 targets
Impact horizon	Cumulative to 2030 (next 5 years)	Achieved by 2030, sustained to 2040
Headline metrics	Deaths & DALYs averted; % ↓ premature NCD mortality by 2030	Deaths & DALYs averted; % ↓ premature NCD mortality by 2040
Interpretation	Value of gains countries have <i>already secured</i>	The <i>full prize</i> of meeting the milestones

From coverage to lives saved: how the model works



One projection engine combines five ingredients

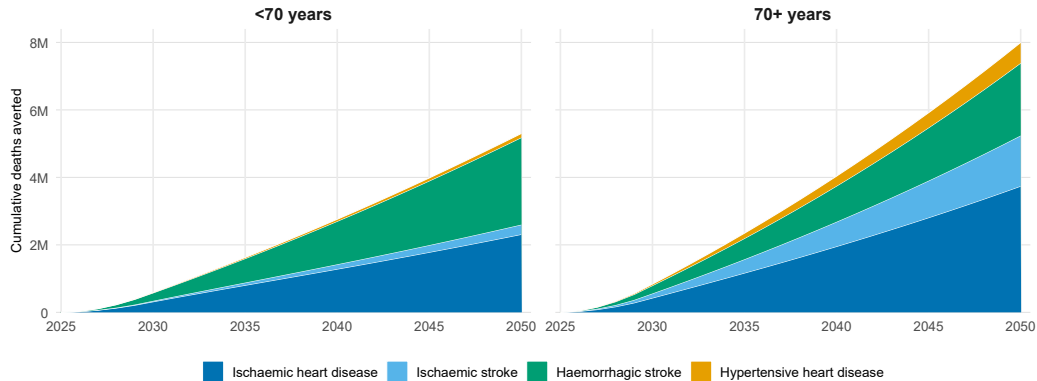
population & demographic projections · country-specific disease burden · intervention coverage · intervention effectiveness · scale-up assumptions

A discrete-time state-transition model projects each country forward; rising milestone coverage lowers disease incidence and death rates. *Details in the appendix.*

WHO South-East Asia Region — combined package (hypertension + statins)

Cumulative deaths averted through 2050

Total 13.3M deaths averted by 2050, vs. the no-additional-scale-up baseline

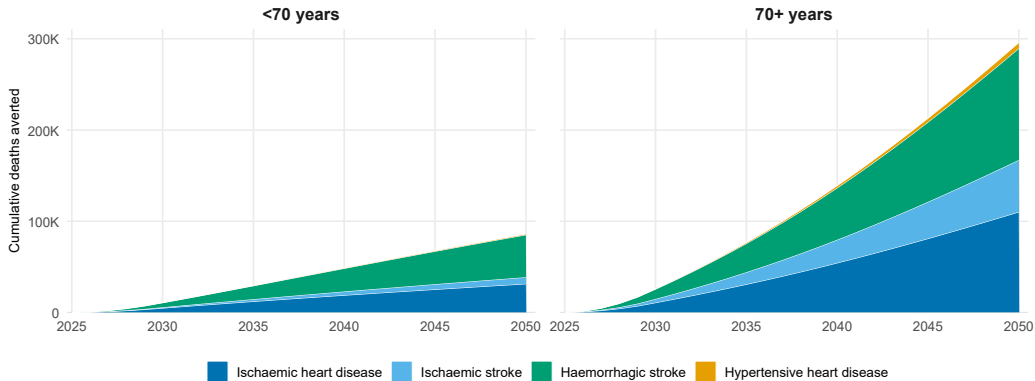


Source: WHO-CVD model output. Values are cumulative relative to the baseline scenario.

Thailand — combined package (hypertension + statins)

Cumulative deaths averted through 2050

Total 382.1K deaths averted by 2050, vs. the no-additional-scale-up baseline



Source: WHO-CVD model output. Values are cumulative relative to the baseline scenario.

Timeline and outputs

- Country-level **and** regional estimates
- By **milestone** and for the **combined** package
- Deaths averted · DALYs averted · % ↓ premature NCD mortality
- Figures & tables ready for **WHO reporting and advocacy**
- Transparent documentation of **data sources & assumptions**

For both estimation exercises

Estimation 1 (progress, 2025→2030) and Estimation 2 (milestones, →2040), with uncertainty where feasible.



Appendix

Appendix 1 — Planned data inflow

Domain / input	Preferred source	Type
SEAHEARTS baselines & targets	WHO SEARO milestone tables (project-provided)	Provided
Population / demography	UN World Population Prospects 2024	External
Cause-specific mortality	WHO GHE / GBD	External
DALYs / disability weights	GBD (YLD, disability weights)	External
Disease incidence & prevalence	GBD	External
Hypertension prev./treatment/control	NCD-RisC; WHO Global report on hypertension	External
Diabetes prev./treatment/control	WHO Global Diabetes Compact; GBD; surveys	External
Tobacco (MPOWER)	WHO MPOWER / GHO tobacco database	External
Sodium (SHAKE)	WHO SHAKE / sodium policy database	External
Trans-fat (REPLACE)	WHO REPLACE / TFA policy database	External
Intervention effect sizes	Trials & meta-analyses (see App. 3–4)	Parameter

Type: *Provided* = WHO SEAHEARTS inputs; *External* = epidemiological / demographic inputs; *Parameter* = effect sizes. Geography = 10 SEAR Member States (country) aggregated to region; years span the 2025 baseline through the 2030/2040 horizons. Sources: UN WPP (United Nations, Department of Economic and Social Affairs, Population Division 2024); GBD (Vollset et al. 2024; Schumacher et al. 2024); NCD-RisC (Zhou et al. 2021); WHO hypertension (World Health Organization 2025); diabetes (Gregg et al. 2023); HEARTS (Moran and Gupta 2023). **Planned, not yet all implemented.**

Appendix 2 — Model engine and burden calculation

Discrete-time state-transition model



- Annual step, single-year ages 20–95, by sex & country
- Causes: IHD, ischaemic & haemorrhagic stroke, hypertensive heart disease
- Background mortality removes population each year
- Each intervention applies multiplicative **effect ratios** to incidence (ε_{IR}) and case-fatality (ε_{CF})
- Coverage scales up **linearly** to target; only *incremental* coverage adds effect (no double-counting)
- Interventions combine **multiplicatively**

As implemented in `scripts/06_run_scenarios_multiple.R`; horizons/targets are being repointed to the SEAHEARTS 2030/2040 exercises.

Deaths → DALYs

$$\text{Deaths averted} = D_{\text{baseline}} - D_{\text{scenario}}.$$

$$\text{DALYs} = \text{YLL} + \text{YLD}:$$

- **YLL** = deaths $\times$ remaining life expectancy (UN WPP)
- **YLD** = prevalent cases $\times$ GBD disability weight

Premature NCD mortality = probability of dying 30–70.

Appendix 3 — Methodology by intervention (as implemented)

Intervention	Baseline exposure	Epidemiologic link / effect size	Status
Hypertension (general)	BP-control rate among hypertensives	Ettehad/BPLTTC RRs by BP bin → ↓ incidence & case-fatality	Implemented
Hypertension (diabetics)	BP control within the diabetic subgroup	Diabetes-specific Ettehad RRs, weighted by diabetes prevalence	Implemented
Sodium (SHAKE)	Mean population salt intake (g/day)	Salt ↓ → SBP ↓ (Filippini) → Ettehad RRs → ↓ incidence	Implemented
Trans-fat (REPLACE)	%E from industrial trans-fat	TFA ↓ → ↓ IHD case-fatality (age-specific RRs)	Implemented
Statins / lipids	Statin coverage (ages ≥40)	Trial RRs (IHD, ischaemic stroke); AF-scaled incidence + case-fatality	Implemented
Tobacco (MPOWER)	MPOWER coverage	—	Planned
Diabetes glycaemic control	Glycaemic-control rate	—	Planned

Scale-up (all implemented modules): linear from the 2025 baseline to the target, effects held thereafter. **Outcomes:** averted incidence/deaths feed the state-transition model → deaths, then YLL+YLD → DALYs. **Combined scenario currently run:** hypertension (general) + hypertension (diabetics) + statins, combined multiplicatively; sodium and trans-fat modules are coded but not yet in the active combined run. **Tobacco (MPOWER) and a dedicated diabetes glycaemic-control → complications module are planned / methodology to be finalized** — the diabetes milestone is presently represented only via BP control in diabetics and statin targeting of the high-risk diabetic population.

Appendix 4 — Key effect-size equations (implemented modules)

Coverage scale-up (all modules)

$$C_t = C_0 + (C_{\text{target}} - C_0) \min\left[\frac{t-t_0+1}{t_1-t_0+1}, 1\right]$$

Hypertension — baseline-adjusted effect (no double-count)

$$E_{\text{adj}}(t) = \frac{E_{\text{trial}}(C_t - C_0)}{1 - E_{\text{trial}} C_0}, \quad \varepsilon_{IR} = \frac{\sum_b IR_b(1 - E_{\text{adj}})P_b}{IR}$$

$$\varepsilon_{CF} = 1 - \rho_{CF} C_{\text{agg}}^{\Delta}$$

Sodium — via blood pressure

$$\Delta\text{SBP} = (2.8r + 1.0(1-r)) \Delta\text{salt}, \quad \varepsilon_{IR} = 1 - (1 - RR) \cdot 0.1 \frac{\Delta\text{SBP}}{\varepsilon_{IR, \text{total}}} = \prod_j \varepsilon_{IR, j}, \quad \varepsilon_{CF, \text{total}} = \prod_j \varepsilon_{CF, j}$$

Trans-fat — IHD case-fatality only

$$E = \frac{\Delta_{\text{TFA}}(RR_a - 1)}{\text{TFA}_0(RR_a - 1) + 1}, \quad \varepsilon_{CF}^{\text{IHD}} = 1 - E$$

Statins (ages ≥ 40 ; IHD & ischaemic stroke)

$$E^{IR} = \frac{AF(1 - RR^{IR})\Delta U}{1 - (1 - RR^{IR})U_0}$$

$$E^{CF} = \frac{(1 - RR^{CF})\Delta U}{1 - (1 - RR^{CF})U_0}$$

Combination

C = coverage; E_{trial} = Ettehad/BPLTTC bin effect (Ettehad et al. 2016); P_b = BP-bin share; ρ_{CF} = cause-specific CF-reduction; r = raised-BP share; RR = relative risk (sodium (Filippini et al. 2021); TFA (Mozaffarian et al. 2006); statins (Fulcher et al. 2015; Baigent et al. 2010)); AF = attributable fraction; U = statin coverage.

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